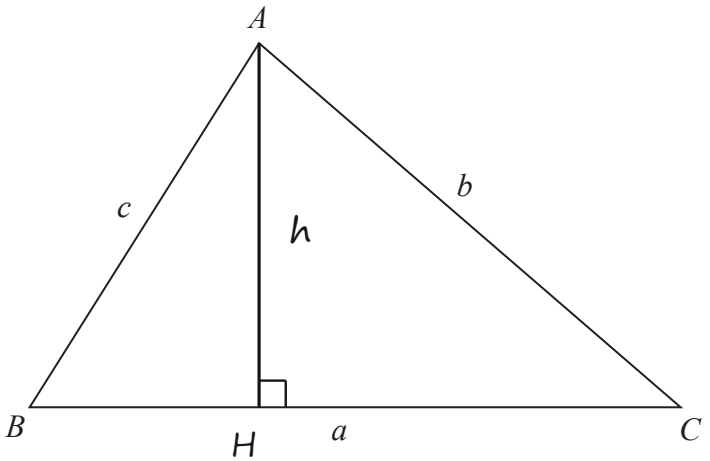
21 The diagram shows an acute-angled triangle ABC.



Prove that area of triangle $ABC = \frac{1}{2}ab\sin C$

Drop AH perpendicular to BC; let AH=h.

In right triangle AHC, sinC = h ÷ b.

1 mark
Use sine for height

Therefore h = b sinC.

Area ABC = ½ × base × height = ½ × a × h.

1 mark
Base × height ÷2

Substitute h: area ABC = ½ab sinC.

1 mark
Complete proof

(Total for Question 21 is 3 marks)